

Getting the Tools

- ☐ Facilitation Skills
- ☐ Active Listening
- ☒ **Problem Solving**
- ☐ Authority and Decision Making
- ☐ The Use of Power and Influence
- ☐ Resolving Conflict in the Team
- ☐ Bringing New Members into the Team



Problem Solving (1/8)

- Signals Your Team Needs Structured Problem Solving
 - Disagreement about what the problem is
 - Failure on the part of the team to identify the goal and the objectives
 - Inability around from problems to solutions to action plans back to problem, etc.
 - Obvious and strong frustration on the part of some members of the team
 - Lack of progress
 - Team members' avoidance of the subject because they believe they can't resolve it

Problem Solving (2/8)

Seven steps of problem solving

- Identifying the problem
- Root Cause Analysis
- Setting objective (*Define Success Criteria*)
- Brainstorming
- Evaluating solutions
- Making a decision
- Developing an action plan
- Evaluating

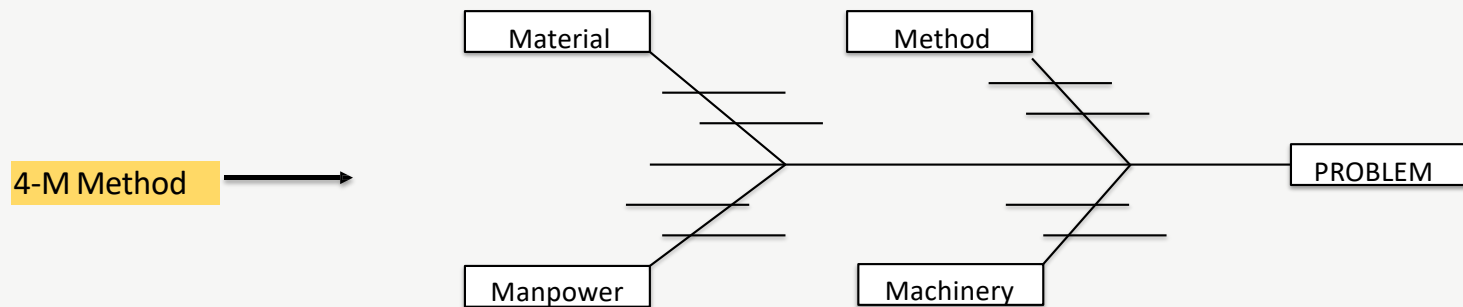
Problem Solving (3/8)

Brainstorming rules

- Everyone tries their best to develop as many new and different alternatives
- Anything goes. There is no such thing as “bad” idea, since it may spur other ones
- Generate as many alternatives as possible
- Use ideas of others to stimulate your own. Try to improve on them or combine them to come up with better approaches
- Encourage each other. Work as a group
- Separate Idea Phase vs Critique Phase
- Equal Voice
- Timebox

Problem Solving (4.1/8)

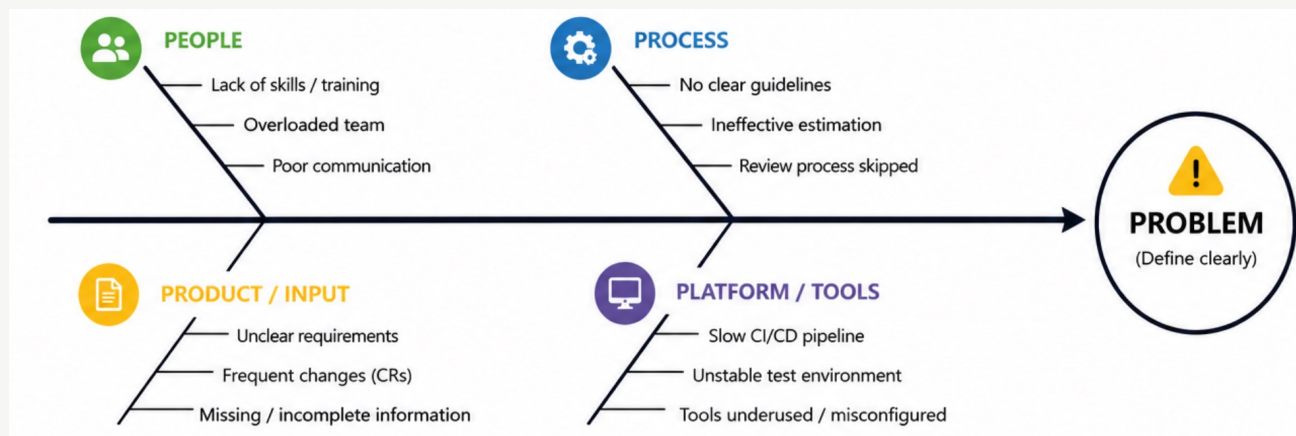
- Techniques to find root causes
 - Cause and effect analysis (Fishbone diagram)



Problem Solving (4.2/8)

- Techniques to find root causes

- Cause and effect analysis (Fishbone diagram) → 4-P Method



Focus on solving the few causes that create the biggest impact.



1. PEOPLE

- Skills & capability gap
- Workload / capacity
- Communication
- Ownership & accountability



2. PROCESS

- Workflow / procedure
- Estimation & planning
- Quality & review process
- Governance & compliance



3. PRODUCT / INPUT

- Requirements clarity
- Change management
- Data / information quality
- Business priorities



4. PLATFORM / TOOLS

- Tools & automation
- Infrastructure / environment
- CI/CD & deployment
- Monitoring & support

Problem Solving (5/8)

- Techniques to prioritize root causes: Pareto analysis (80/20 rule)
 - Identify the 20% root causes that caused 80% of the problem
 - Put the first effort upon the 20% of the tasks that can bring in 80% satisfaction of the customer
 - Identify the place where we can spend 20% of budget to address 80% of the risk

Pareto for Engineering Excellence

1. Which 20% blockers slow delivery most?
2. Which 20% defects hurt clients most?
3. Which 20% process gaps create rework?
4. Which 20% talents create 80% leverage?

Problem Solving (6/8)

- Techniques to stimulate lateral thinking: Six hat thinking

White

Cost, license, productivity data?

Red

Dev lo bī replace?

Black

Security leak? IP risk?

Yellow

Faster coding, learning boost

Green

Pilot only for selected teams?

Blue

Decision today: run 60-day pilot



Yellow hat: good or positive. The sun is shining down

→ optimism with logic



Red hat: emotions, or, how we feel about it. People red faces often give away their emotional state, gut reaction → emotional truth



White hat: deals with the facts. Plain as a white paper → logic mode



Green hat: symbolizes new and different ideas. New life, creative → innovation mode



Blue hat: the thinking process itself. The sky sits over us. "what should we do next?" → meeting leader hat



Black hat: possible cautions or concerns. Provide judgments. Focus upon negative aspects → risk management

Problem Solving (7/8)

- Techniques to stimulate creative thinking SCAMPER
 - **Substitute** – components, materials, people
 - **Combine** – mix, combine with other assemblies or services, integrate
 - **Adapt** – alter, change function, use part of another element
 - **Modify** – magnify, reduce, minimize in scale, change shape, modify attributes (e.g. colour)
 - **Put** to another use
 - **Eliminate** – remove elements, simplify, reduce to core functionality
 - **Reverse** – turn inside out or upside down

Problem Solving (8/8)

- Developing commitment from others outside the team through involvement
 - Anticipating problem and laying groundwork
 - Focusing team members on involving others
- Follow-up and team learning